

Exp No:15

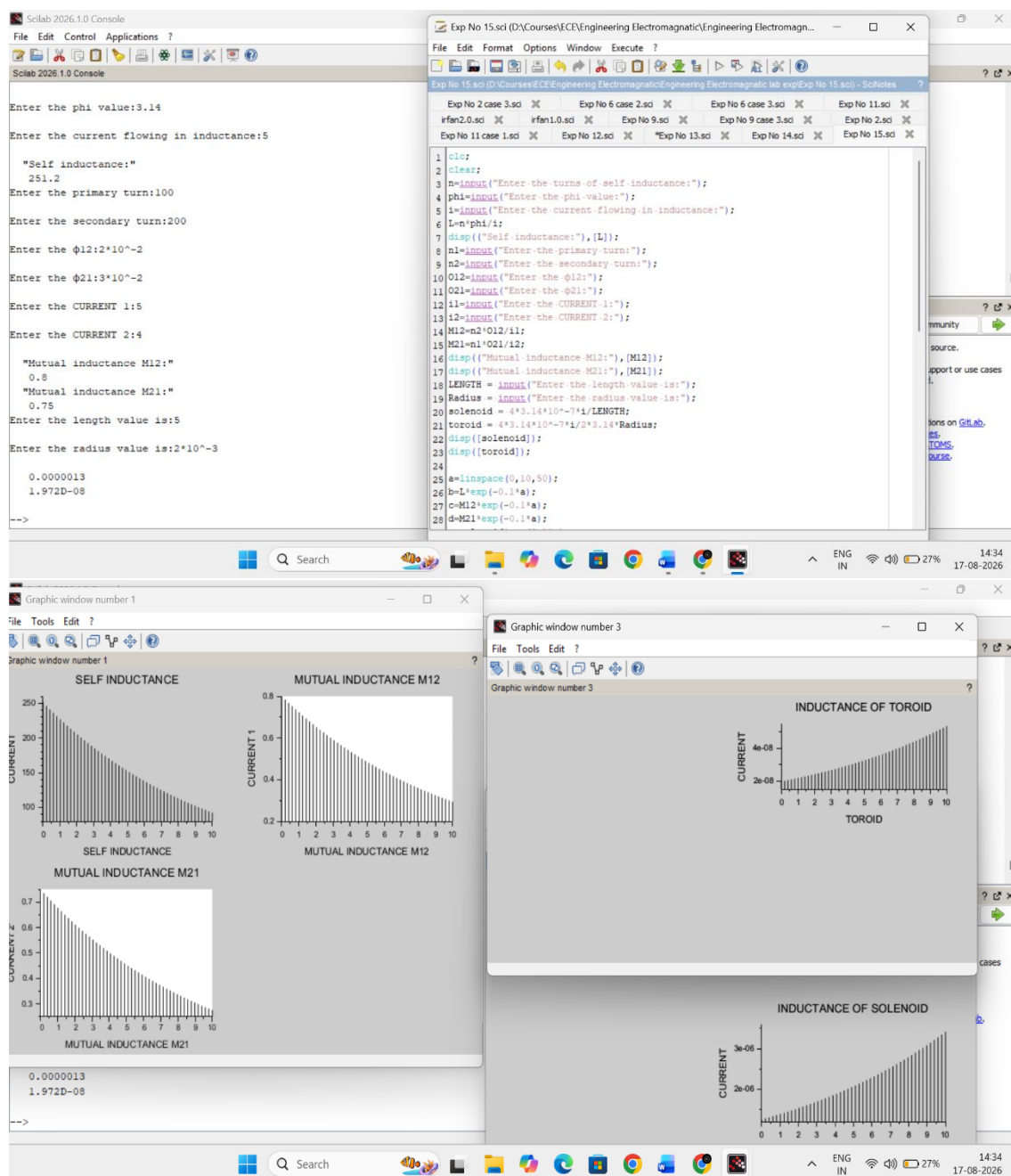
Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1



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no of turns in self inductor
 $N = 400$

$$i = 5$$

$$n_1 = 100$$

$$n_2 = 200$$

length =

$$\text{length} = 5 \text{ cm}$$

$$i_1 = 5 \text{ A}$$

$$i_2 = 4 \text{ A}$$

$$\phi_{12} = 2 \times 10^{-2}$$

$$\phi_{21} = 3 \times 10^{-2}$$

$$\text{radius} = 2 \text{ cm}$$

~~Example~~

Determine self inductance and MI (i) solenoid and (ii) toroid with different length and circumferential area

Self Inductance

$$L = n \phi_i$$

$$n = \frac{251.2}{3.14 \times 5} = 4$$

$$L = 251.2 \text{ H}$$

Mutual Inductance

$$M_{12} = \frac{N_2 \phi_{12}}{I_1}$$

$$M_{12} = \frac{200(2 \times 10^{-2})}{4}$$

$$= 1/5$$

$$M_{12} = 0.3 \text{ H}$$

$$M_{21} = \frac{N_1 \phi_{21}}{I_2}$$

$$M_{21} = \frac{100(3 \times 10^{-2})}{4}$$

$$= 3/4$$

$$M_{21} = 0.75 \text{ H}$$

Solenoid

$$L = \frac{\mu \pi \times 10^{-7} I}{l}$$

$$= \frac{4(3.14 \times 10^{-7})(5)}{5}$$

$$= 1.256 \times 10^{-6}$$

$$L = 1.256 \times 10^{-6}$$

$$L = 0.0000012$$

Toroid

$$L = \frac{\mu_0 I}{2\pi r}$$

$$L = \frac{(4\pi \times 10^{-7})(5)}{2\pi(2 \times 10^{-3})} = \frac{20 \times 10^{-7}}{4 \times 10^{-3}}$$

$$= 5 \times 10^{-4}$$

$$L = 5 \times 10^{-4}$$

$$L = \frac{4 \times 3.14 \times 10^{-7} \times 5}{2 \times 3.14 \times 2 \times 10^{-3}}$$

$$= \frac{6.28 \times 10^{-6}}{2 \times 3.14 \times 2 \times 10^{-3}} = 1.972 \times 10^{-3}$$

$$L = 1.972 \times 10^{-3}$$

Result:

Thus, determination and plotting of Inductance, Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area

Are performed and verified successfully.